Multilevel Joint Regularizations Meet Dual Self-Representations: An Innovative Deep Subspace Clustering Framework for Hyperspectral Images

Han Zhai, *Member, IEEE*, and Jun Yu, *Student Member, IEEE*

***Abstract*—Hyperspectral image (HSI) Clustering is a long-standing difficulty due to the huge spectral variability and significant nonlinearity. Deep subspace clustering (DSC) has presented incomparable advantages over traditional subspace clustering by learning robust features and nonlinear affinities. However, due to the neglect of hierarchical complementarity by solely using the deepest features and insufficient exploration of pixel relationships through single self-representation, the existing DSC methods suffered from degraded performance. To address these issues, this article proposed an innovative clustering framework named dual self-representation-based DSC with multilevel joint regularizations (DDSC-MJR) for HSIs. On the one hand, a dual self-representation (DSR) module was developed to fully excavate the intrinsic affinities between pixels by consecutive self-expressions and affinity learning. On the other hand, both deep-seated and shallow features induced by a well-designed convolutional autoencoder (CAE) were exploited in dual self-expressions, with multilevel joint regularizations (MJRs) used to promote their coefficient matrices to share a similar structural pattern. To fully leverage the advantages of multilevel self-expressions, a weighted residual fusion (WRF) module was developed to adaptively fuse the learned coefficient matrices to create a more indicative and clustering-friendly adjacency matrix. Ultimately, a multifaceted joint loss involving data reconstruction, DSRs, MJRs, and graph regularization (GR) was put forward to simultaneously constrain each part of the network in the training process. The extensive experimental results on three public HSI datasets clearly demonstrate the large superiority of DDSC-MJR to the other state-of-the-art (SOTA) arts and indicate that it can be a promising tool for real applications.**

***Index Terms*—Deep subspace clustering (DSC), dual self-representation (DSR) module, hyperspectral images (HSIs), multilevel joint regularizations (MJRs), weighted residual fusion (WRF) module.**

I. Introduction

HYPERSPECTRAL images (HSIs) record fine-grained spectra of various ground objects through a large number of narrow electromagnetic wave bands while providing wealthy spatial contents for the interested areas to enable refined information extraction [1], [2], [3]. Clustering and classification are regarded as two basic techniques for HSI interpretation. Comparatively, limited by the labor-intensive and difficult collection of training samples, classification is less popular and attractive in practice [4], [5]. Clustering, which is also known as "unsupervised classification," aims at grouping similar pixels while separating dissimilar pixels by data mining without any training samples, which has widespread applications, such as geology survey [6], fine agriculture [7], smart city [8], and military defense [9]. However, HSI clustering is very challenging and susceptible to the high-dimensionality and nonlinearity of the data as well as serious variations in spectra and appearance of ground objects.

In the past decades, numerous traditional clustering methods have been proposed for HSIs, which mainly include five kinds according to their principles [10], [11]. The first kind is the centroid-based models, which are the earliest representatives and cluster pixels by updating centroids and the regarding pixels in an iterative way to minimize the partition error, like k-means [12], fuzzy c means [13] and their diverse variants [14], [15]. The second category is the density-based approaches, which partition pixels according to densities and relative distances, such as density peak-based clustering [16] and diffusion-based clustering [17]. These methods adapt well to cluster shapes and are robust to disturbances. The third class is the probability-based models, which simulate pixel distribution with specific probability density functions and separate them according to certain probability rules, with the Gaussian mixture model [18] and expectation maximization as typical examples [19]. The fourth one is the biology-inspired methods, which introduce advanced biological models to simulate the complex structure of HSIs to induce more accurate clustering results, including immune network [20] and deoxyribonucleic acid (DNA) network [21]. The last one is the graph-based models, which construct affinity graphs based on the original image features to describe pixel relationships and generate clusters by graph cutting, such as sequential spectral clustering [22] and minimax bridge clustering [23]. Moreover, a set of anchor graph-based methods and co-clustering frameworks were developed to improve model scalability and efficiency for use with large HSIs [24], [25], [26]. However, the complex structure and varied spectra of HSIs often violate the native assumptions of these methods, leading to limited clustering performance.

To alleviate these problems, scholars devote to developing more effective approaches to reveal the intrinsic pattern

Received 18 September 2025; revised 5 December 2025; accepted 7 December 2025. Date of publication 11 December 2025; date of current version 9 January 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 42271386 and Grant 42001313, in part by the National Science and Technology Basic Resource Investigation Project under Grant 2019FY202503, in part by the National Earth Observation Data Center Project under Grant NODAOP2025005, and in part by the Key Laboratory of Monitoring, Evaluation and Early Warning of Territorial Spatial Planning Implementation, Ministry of Natural Resources Project under Grant LMEE-KF2025013. *(Corresponding author: Han Zhai.)*

The authors are with the School of Geography and Information Engineering, China University of Geosciences, Wuhan, Hubei 430074, China (e-mail: zhaihan@cug.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3643295

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of high-dimensional data, which gives birth to subspace clustering. Subspace clustering learns underlying pixel affinities in a union of low-dimensional subspaces by data self-representation and applies spectral clustering to the adjacency matrix to generate a meaningful clustering result, which adapts well to the complex structure and varied spectra of HSIs [10], [27]. Since the successful trail of spectral-spatial sparse subspace clustering (S^4C) for HSIs [28], [29], extensive follow-up works have been carried out [30], [31], [32], [33], [34], [35], [36], [37], [38], which greatly promote the development of subspace clustering and make it to be a recent area of focus. Among them, the typical representatives are SSC [29], [30], [31], [32], [33], [34] and low-rank subspace clustering (LRSC) models [39], [40], which employ HSI data itself as the dictionary and explore the sparsity or low-rank property of self-expression coefficients by enforcing pixels to be expressed by a few same categorial atoms or reconstructed by all atoms in the dictionary to induce robust and favorable pixel affinities. Based on these, various enhanced versions have been put forward to learn more precise pixel affinities from different perspectives, such as kernelized models [41], [42], superpixel models [31], [43], multiview models [44], [45], dictionary learning-based models [46], multiobjective optimization [47], structural prior constraints [48], [49], high-order information mining [50], [51], and graph regularizations (GRs) [51], [52], [53]. Moreover, considering of the prohibitive complexity caused by self-dictionary expression, a set of scalable subspace clustering models were developed for large HSIs to further improve model practicability [54], [55], [56]. For instance, by constructing a compact dictionary for efficient self-expression with sparse low-rank constraints and nonlocal means regularization used to explore the structural properties and self-similarity of HSIs, a competitive sketched subspace clustering model was proposed, which achieved impressive results and presented a large practical potential [54]. In addition, by transforming the large-scale matrix sparse optimization problem into a pixel-wise sparse reconstruction problem, Zhai et al. [55] proposed efficient sparse coding and joint sparse coding clustering models, which delivered superior performance to prior arts. Although remarkable accomplishments have been achieved, these methods are essentially shallow linear models with limited discriminability and large difficulty in processing the obvious nonlinearity of HSIs. In spite of some kernelized models' attempt to relieve data nonlinearity by feature mapping, the hand-crafted kernels commonly have poor generalizability and weak adaptability to complex scenes.

The rapid advancement of deep learning has opened up new avenues for HSI clustering, whereas its noticeable feature extraction and nonlinear expression abilities provide a feasible solution to the above problems. Consequently, extensive deep clustering methods have been developed by combining deep neural networks (DNNs) with traditional clustering models to learn favorable and clustering-friendly high-level features [57], [58], [59]. For example, by jointly optimizing an autoencoder (AE) and k-means, deep clustering network [60] and deep embedding clustering [61] significantly outperformed traditional clustering methods. In addition, by using a pretrained or randomly initialized convolutional neural network (CNN) as the primary feature extractor and iteratively optimizing clusters and network parameters, the separate CNN-based clustering models yielded impressive results [62]. Moreover, various generative adversarial network (GAN) or variational AE-based deep clustering methods were put forward to learn the latent data distribution for sample generation, thus obtaining higher accuracy [63]. By introducing DNNs to spectral clustering to simultaneously conduct embedding learning and graph cutting, more accurate clustering results can be induced [58]. Besides, by fully leveraging deep information generation and degradation model information through unsupervised learning, innovative superresolution models were developed for HSIs, with competitive performance obtained [64], [65]. In addition, based on unsupervised representation learning, superresolution was effectively extended to unpaired restoration of images and point clouds [66]. All these achievements indicate the large potential of deep learning in unsupervised tasks. However, the above methods lack of strict descriptions of the intrinsic adjacency relationships between pixels, resulting in restricted clustering accuracy.

On this account, deep subspace clustering (DSC) effectively overcomes this shortcoming by taking full advantage of subspace clustering and deep learning. Benefiting from robust nonlinear feature extraction, DSC can capture more precise pixel affinities to yield much better results than traditional subspace clustering, which has attracted increasing attention with many competitive models developed for HSIs [67], [68], [69]. For instance, with 3-D convolutional AEs (CAEs) used for spectral-spatial feature extraction, a Laplacian regularized DSC [70] and an attention enhanced DSC were proposed [71], which obtain impressive results. By using multiscale inputs to enrich encoding features, a competitive spectral-spatial DSC method was put forward [72]. By integrating spectral clustering into the DSC framework, an end-to-end self-supervised model was developed, which delivers noteworthy performance [73]. In addition, on the basis of graph convolutional regularization (GCR), a residual DSC was developed [74]. Then, several enhanced models were put forward to further improve the clustering accuracy by introducing mutual information [75], hypergraph structure [76], and multiview features [77], respectively. Given serious interference from the background, a robust low-rank graph convolutional DSC was proposed by integrating spectral and spatial features learned by 1-D and 2-D CAEs to construct a robust adjacency matrix, thus yielding higher accuracy [78]. Furthermore, based on contrastive learning, a dual-branch CNN-based DSC was proposed, which facilitates efficient difference learning of positive and negative sample pairs [79]. Then, a superpixel neighborhood and multiview enhanced contrastive learning models were developed, which adapt well to large HSIs [69], [80]. Moreover, with the popularity of vision transformer, some transformer-based DSC models have been developed to better mine global expressions, thereby inducing more competitive performance [81], [82]. Besides, a novel DSC model was developed by embedding an unfolding alternative direction of multipliers method (ADMM) into subspace clustering, thus deeply optimizing pixel affinities [83].

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However, in spite of these advances, the existing DSC methods still encounter some common bottlenecks. First, current networks only focus on the deep-seated feature expressions and solely use the deepest feature for self-representation learning, but ignore the importance and complementarity of shallow features, which inevitably leads to certain discriminability loss. Second, current models learn pixel affinities by single self-representation, which is susceptible to the expressivity and robustness of features and prone to missing vague but important relationships. As a result, the induced adjacency matrix cannot precisely recover the underlying partition pattern of HSIs.

In view of these, in this article, an innovative dual self-representation (DSR)-based DSC network with multilevel joint regularizations (DDSC-MJR) was proposed for HSIs, which establishes a novel DSC framework to improve the clustering performance in complex scenes. In the DDSC-MJR model, an effective channel attention embedded CAE was constructed to deeply mine expressive and robust semantic features in an unsupervised manner, with channel attentions used for feature refinement and skip connections employed for feature cross-layer transmission to assist data reconstruction and reduce detail loss. Different from traditional DSC models, a DSR module was designed between the encoder and decoder for successive self-expression learning, with both shallow and deep features exploited to integrate the advantages of multilevel expressions. The learned coefficient matrices were adaptively fused by a weighted residual fusion (WRF) module to induce a high-quality adjacency matrix for spectral clustering to generate more accurate clustering results. In addition, an additional GR was imposed on the fused coefficient matrix to mine the latent geometry in the non-Euclidean space. The contributions mainly involve fourfold as follows.

1) An innovative DSC framework termed DDSC-MJR was proposed for HSIs based on DSRs and multilevel joint regularizations (MJRs). Differing from current DSC models, DDSC-MJR explores the latent structural pattern of HSIs by consecutive pixel affinity learning with the complementarity of multilevel features fully explored. To simultaneously constrain each part of the model, an effective multifaceted joint loss function composed of data reconstruction loss, DSR loss, MJR loss, and graph loss was developed to train the network from scratch. All of these significantly improve the robustness of the model to complex scenes.
2) To address the shortcoming of single self-representation, an efficacy DSR module was developed, which pays full attention to the value of the reconstructed feature and excites its potential by another round of self-expression to make up for adjacency loss in the first self-expression. Different from the existing double self-representations [84], it uses the reconstructed feature to conduct another same-domain self-representation, rather than employs the learned coefficients as feature proxies to perform cross-domain self-representation, thereby avoiding expression bias caused by the differences in statistical characteristics of different domains. Via hierarchical self-representation learning, the intrinsic relationships between pixels can be fully mined, which captures more precise pixel affinities and directly contributes to the final clustering accuracy.
3) To fully leverage the advantages of multilevel features in joint self-representations to compensate for nonnegligible discriminability loss caused by unilateral preference of the deepest feature in the existing works, forceful MJRs were put forward based on similarity assessment and contrastive learning, to promote consistent expression of deep and shallow features and enforce their coefficient matrices to share a similar structural pattern, thereby improving the discriminability of the model. All regularizations are applied to coefficient matrices, thus realizing collaborative assistance and mutual promotion in the self-representation domain.
4) To make the best of the complementarity contained in different coefficient matrices, a valid WRF module was designed to adaptively fuse the learned coefficient matrices to generate a complete and clustering-friendly adjacency matrix. It learns the regarding weight for each coefficient matrix based on a channel attention and reserves detailed pixel relationships through a residual connection, which precisely integrates the differentiated information and minimizes information loss as much as possible. As a result, clear gains can be achieved in clustering accuracy.

The outline of this article is as follows. Section II briefly introduces subspace clustering and DSC models. Section III intimately insights the structure and scheme of DDSC-MJR. Experiments and detailed analysis are presented in Section IV. Section V summarizes this article and gives future lines.

II. Related Work

A. Subspace Clustering for HSIs

The basic idea of subspace clustering is to seek a linear reconstruction for each pixel with other pixels residing in a union of subspaces under certain prior constraints, thereby capturing the latent relationships between pixels. Given a HSI data $\mathbf{H} = [\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_n] \in \mathbb{R}^{d\times n}$, where d and n denote the channel number and pixel number, with data itself as the dictionary, the self-representation problem can be formulated as in the following equation:

$$\mathbf{H} = \mathbf{HC} + \mathbf{E} \tag{1}$$

where $\mathbf{C} \in \mathbb{R}^{n\times n}$ denotes the associated coefficient matrix that records the underlying pixel affinities, and $\mathbf{E} \in \mathbb{R}^{d\times n}$ stands for the expression error matrix.

It should be noted that (1) may have infinite solutions. Therefore, to obtain a meaningful and stable solution, some prior constraints should be imposed on $\mathbf{C}$ to reduce the solution space [85], [86], as in the following equation:

$$argmin_{\mathbf{C}} \|\mathbf{C}\|_p + \lambda \|\mathbf{H} - \mathbf{HC}\|_F^2 \tag{2}$$

where $\|\cdot\|_p$ denotes the p-norm, which restricts the structure of $\mathbf{C}$, with $\|\cdot\|_1, \|\cdot\|_*$, and $\|\cdot\|_F^2$ commonly used, referring to

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SSC, LRSC, and collaborative representation clustering (CRC) models [10], respectively, and λ is a tradeoff parameter. It should be noted that in the SSC models, another constraint $\text{diag}(\mathbf{C}) = \mathbf{0}$ should be added to avoid each pixel being expressed only by itself. By promoting the sparsity or row-rank property of $\mathbf{C}$, SSC and LRSC models can explore the local and global structural information of HSIs, respectively, which leads to robust performance [29], [30], [39], [40]. By minimizing the energy of $\mathbf{C}$, CRC models greatly improve the clustering efficiency [46].

Given the absence of spatial information in (2), by jointly modeling the spectral-spatial duality of HSIs as in (3), much higher accuracy can be derived:

$$argmin_{\mathbf{C}} \|\mathbf{C}\|_p + \lambda \|\mathbf{H} - \mathbf{H}\mathbf{C}\|_F^2 + \tau \Gamma(\mathbf{C}) \tag{3}$$

where $\Gamma(\cdot)$ denotes certain spatial regularization, and τ is a tradeoff parameter between the spectral and spatial terms.

The above problems are generally solved by the ADMM method. Subsequently, by applying spectral clustering to the adjacent matrix induced by the learned coefficient matrix as in (4), the final clustering results can be obtained [29], [30].

$$\mathbf{W} = |\mathbf{C}| + |\mathbf{C}^T| \tag{4}$$

Based on these, generous enhanced subspace clustering models have been developed to further improve the clustering performance from different perspectives. Specifically, to reduce the systematic error caused by linear approximation of nonlinear structure of HSIs, kernel subspace clustering models were proposed to translate linear inseparable problem into linear separable problem by feature mapping with customized kernels, which deliver impressive results [41], [42]. By treating homogenous objects as basic units, superpixel-based subspace clustering models presented large advantages in both accuracy and efficiency [31], [43]. Given the complementarity of features from different domains, various multiview subspace clustering models were developed by extending (2) to multifeature joint self-expression problems, thereby obtaining higher accuracy [44], [45]. By learning more indicative and expressive dictionaries for subspace clustering, competitive performance was derived [46]. Moreover, by introducing reasonable structural priors to guide self-expressions [48], [49], and exploring high-order relationships with hypergraphs [50], [51], clustering accuracy was further improved. In recent years, a set of works have shown the efficacy of GCRs in geometric information mining in the non-Euclidean space [51], [52], [53]. Overall, subspace clustering has made significant progress, but is still limited by its inherent shallow and linear properties.

B. DSC for HSIs

DSC inherits the advantages of subspace clustering while upgrading its ability in deep feature extraction and nonlinear affinity mining with the help of DNNs, which is more in line with the characteristics of HSIs and presents incomparable advantages over traditional subspace clustering [67], [68], [69]. The crux of DSC is to learn discriminative feature expressions and jointly optimize feature extraction and self-expression.

Generally, a DSC network consists of two parts, i.e., an AE composed of a fully connected network (FCN) or CNN for feature extraction and a fully connected (FC) layer for self-representation [70], [71], [72]. By minimizing data reconstruction error, abstract and robust deep-seated features can be learned in an unsupervised manner. Specifically, given a parameterized AE with encoder and decoder parameters denoted as Θ_e and Θ_d, a general loss function of DSC can be formulated as in the following equation:

$$\ell(\Theta, \mathbf{C}) = \alpha \|\mathbf{H} - \hat{\mathbf{H}}\|_F^2 + \beta \|\mathbf{Z}_{\Theta_e} - \mathbf{Z}_{\Theta_e}\mathbf{C}\|_F^2 + \gamma \|\mathbf{C}\|_p \tag{5}$$

where $\hat{\mathbf{H}} \in \mathbb{R}^{d \times n}$ denotes the reconstructed data by AE, $\mathbf{Z}_{\Theta_e}$ stands for the learned feature by encoder, α, β and γ represent regularization parameters to balance different terms. Through deep feature engineering, nonlinear feature expressions can be induced, which align better with the assumption of subspace clustering and contribute to more precise self-expression and pixel affinity learning. Analog to subspace clustering, by applying spectral clustering to the adjacency matrix built as in (4), reasonable clustering results can be obtained [70], [71], [72].

Nowadays, DSC has become more and more popular, with many excellent models developed for HSIs from diverse aspects. Specifically, by building a 3-D CAE to simultaneously learn spectral-spatial feature expressions and introducing the Laplacian regularization to restrict the internal structure of the coefficient matrix, impressive results were obtained [70]. On this basis, by embedding attentions into 3-D CAE to learn more favorable features for self-expression, higher accuracy can be induced [71]. By learning spectral and spatial features through 1-D and 2-D CAEs separately and fusing their complementarity to induce more precise pixel affinities, state-of-the-art (SOTA) performance was obtained [67], [78]. In addition, by exploring the potential of GCRs, a set of advanced DSC models has been put forward. For example, a residual DSC was put forward by adding residual connections to the CAE to promote gradient propagation [74]. Based on this, a mutual information DSC [75] and hypergraph structure AE (HyAE) [76] were put forward to further improve the clustering performance, by learning more intuitive feature space regarding to semantics and mining more favorable structural features with hypergraph, respectively. Moreover, by introducing contrastive learning to enlarge the difference between positive and negative sample pairs, a novel DSC framework was proposed, which delivers very competitive performance [79]. Besides, by exploring the complementarity of multiple features from different domains, many multiview DSC models were developed to induce higher accuracy [69], [77]. Although DSC has achieved noticeable successes, current methods still suffer from limited clustering performance because of insufficient exploration of hierarchical complementarity and intrinsic pixel adjacency relationships.

III. Dual Self-representation based Deep Subspace Clustering Network with Multilevel Joint Regularizations

In this section, an innovative DSC framework named DDSC-MJR is proposed for HSIs, which fully exploits the hidden pixel affinities through successive self-expressions and

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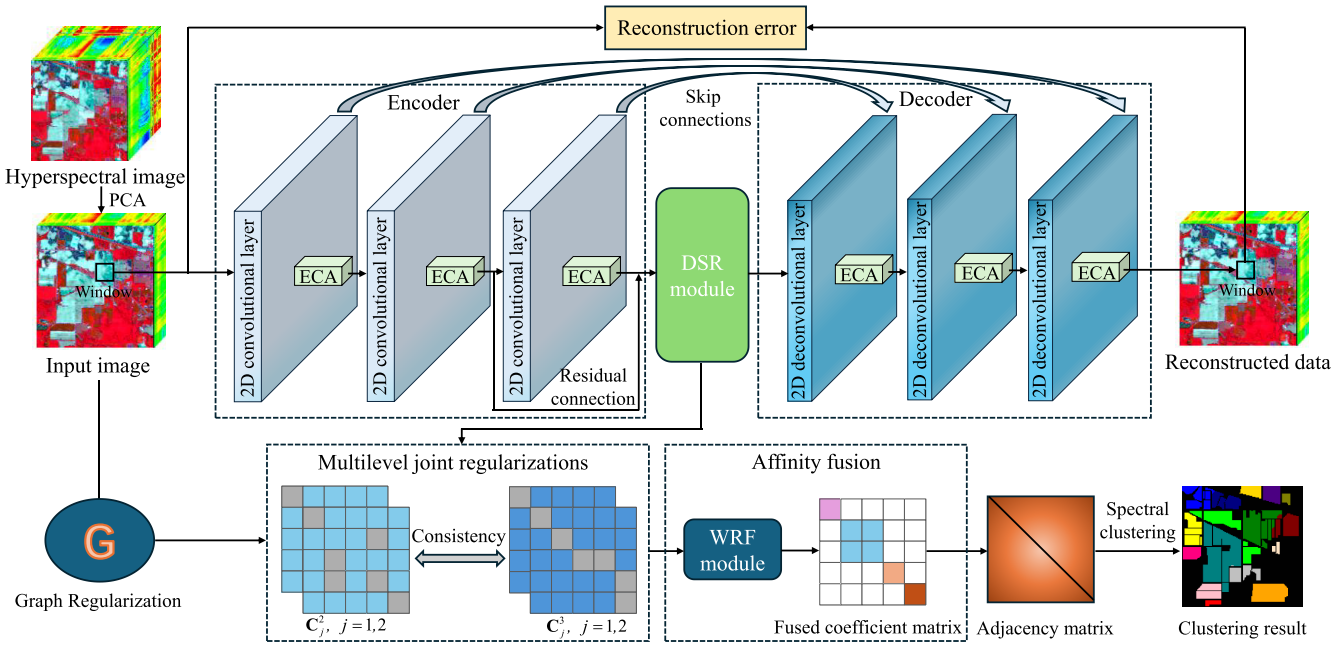


Fig. 1. Overall architecture of the DDSC-MJR model. A centralized window is opened for each pixel of the reduced dimensional image through PCA to generate the input for the ECA embedded CAE. Then, the middle-level and high-level encoding features are fed into the DSR module for consecutive self-expression learning, with MJRs constructed to promote consistent expression. The GR is introduced to enhance model robustness. The learned coefficient matrices are adaptively fused through the WRF module, with the result used to construct the adjacency matrix. Finally, through spectral clustering, the final clustering result can be obtained.

integrates hierarchical complementarity by MJRs, to promote robust clustering performance in complex scenes.

A. Overall Architecture

The overall framework of the proposed DDSC-MJR model is depicted in Fig. 1. As illustrated, with a symmetrical deep CAE used as the skeleton of the network, DDSC-MJR mainly includes five parts. The first part is the channel attention embedded encoder, which is responsible for discriminative feature mining. The second part is the decoder, which is used for data reconstruction and regularizing feature extraction in the encoding stage. The third part is the DSR module, which is embedded between the encoder and decoder and responses for pixel affinity mining through two consecutive self-expressions. The fourth part is the WRF module, which is set up to adaptively fuse coefficient matrices learned by the DSR module. Apart from these four trainable parts, a spectral clustering part is deployed after the WRF module, which is a separatable part and is used to generate the final clustering result based on the adjacency matrix induced by the fused coefficient matrix.

To reduce band redundancy of HSIs and computational complexity, a common strategy is used to reduce the channel number to five with principal component analysis (PCA) [74], [75], [76], [77]. Subsequently, a centralized window at a size of $s \times s$ is opened for each pixel to generate the input patch for the CAE. Hence, the input data $\mathbf{X}$ can be expressed as a set of patches, i.e., $\mathbf{X} = (\mathbf{X}_1, \mathbf{X}_2, \cdots \mathbf{X}_n) \in \mathbb{R}^{s\times s\times 5\times n}$. Specifically, the encoder consists of three consecutive 2-D convolutional layers to hierarchically extract discriminative feature expressions, with the encoding feature of each layer denoted as $\mathbf{Z}_i^{\Theta_e} \in \mathbb{R}^{s\times s\times k_i\times n}, i = 1,2,3$, where k_i denotes the kernel number. Herein, 3×3 convolutions are used for local content learning, with the kernel number set as 24, 24, and 32 for the three layers, respectively. To keep feature size unchanged, the same padding operation is employed for each layer, with batch normalization (BN) and rectified linear unit (ReLU) also used. To highlight important information and explore the interactions between different feature patterns, an efficient channel attention (ECA) [87] is deployed in each layer to further refine the convolutional features.

To ensure that the encoding features capture the key latent information of the input data, an accompanying decoder is built to reconstruct the data and impose necessary constraints on the encoding stage. Herein, a symmetrical structure is adopted, with three successive deconvolutional layers constituted in the decoder. The decoding feature of each layer is denoted as $\mathbf{D}_i^{\Theta_d} \in \mathbb{R}^{s\times s\times k_i\times n}, i = 1,2,3$. Similarly, 2-D deconvolutions with a kernel size of 3×3 are used for the three layers, and the kernel number is set as 32, 24, and 24, respectively, with an ECA deployed in each layer for feature refinement. In order to make full use of multilevel features to better reconstruct the data, skip connections are deployed between the encoding and decoding layers for feature transmission, with them fused by an element-wise addition. By minimizing the reconstruction error as in (6), an elaborate reconstructed data $\hat{\mathbf{X}} \in \mathbb{R}^{s\times s\times 5\times n}$ can be obtained.

$$loss_{CAE} = \alpha\|\mathbf{X} - \hat{\mathbf{X}}\|_F^2 \tag{6}$$

For the DSR module, with the flattened middle-level and high-level encoding features $\tilde{\mathbf{Z}}_2^{\Theta_e}$ and $\tilde{\mathbf{Z}}_3^{\Theta_e}$ as the input, the hidden pixel affinities are fully exploited through consecutive

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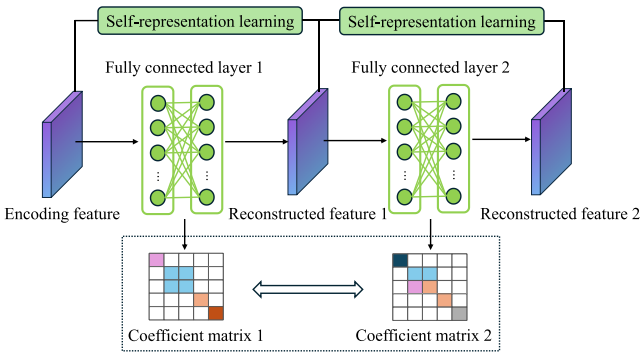


Fig. 2. Structure of the DSR module.

self-representation learning. Consequently, four informative coefficient matrices, i.e., $\mathbf{C}_1^2, \mathbf{C}_2^2, \mathbf{C}_1^3$, and $\mathbf{C}_2^3 \in \mathbb{R}^{n\times n}$ are derived. Subsequently, these four coefficient matrices are fed into the WRF module for fusion to generate a more complete and clustering-friendly coefficient matrix $\mathbf{C} \in \mathbb{R}^{n\times n}$, which well recovers the intrinsic relationships between pixels.

Based on the fused coefficient matrix, an adjacency matrix $\mathbf{W}$ can be constructed as in (4). Then, through spectral clustering, the final clustering result can be yielded.

B. Dual Self-Representation Module

Given the complexity of the internal structure of HSIs and variability of spectra and appearance of ground objects, single self-expression learning is commonly not enough for intricate pixel affinity mining, with vague but vital relationships easily missed, resulting in a congenitally deficient coefficient matrix, which limits the clustering performance to a large extent. Based on this fact, a natural idea is to extend the single self-expression to consecutive self-expressions, to optimize pixel affinities and supplement for omitted pixel relationships, thus improving the quality and expressivity of the coefficient matrix.

On this account, an effective DSR module is constructed to fully exploit the intrinsic pixel relationships by two consecutive self-representations to learn more expressive and informative coefficient matrices, with its structure described in Fig. 2. The DSR module consists of two FC layers, with the first FC layer referring to a conventional self-expression based on the deepest encoding feature as in current DSC models and the second FC layer regarding to a supplementary self-expression based on the reconstructed feature, which can be formulated as follows:

$$argmin_{\mathbf{C}_1}\|\mathbf{C}_1\|_F^2 \; s.t. \; \tilde{\mathbf{Z}}_3^{\Theta_e} = \tilde{\mathbf{Z}}_3^{\Theta_e}\mathbf{C}_1 + \mathbf{E}_1 \tag{7}$$

$$argmin_{\mathbf{C}_2}\|\mathbf{C}_2\|_F^2 \; s.t. \; \tilde{\mathbf{Z}}_3^{\Theta_{FC1}} = \tilde{\mathbf{Z}}_3^{\Theta_{FC1}}\mathbf{C}_2 + \mathbf{E}_2 \tag{8}$$

where $\tilde{\mathbf{Z}}_3^{\Theta_e}$ denotes the flattened feature of $\mathbf{Z}_3^{\Theta_e}$, $\tilde{\mathbf{Z}}_3^{\Theta_{FC1}} \in \mathbb{R}^{32s^2\times n}$ and $\tilde{\mathbf{Z}}_3^{\Theta_{FC2}} \in \mathbb{R}^{32s^2\times n}$ stand for the reconstructed features by the first and second FC layers, with Θ_{FC1} and Θ_{FC2} being their parameters, which are expressed as $\tilde{\mathbf{Z}}_3^{\Theta_{FC1}} = \tilde{\mathbf{Z}}_3^{\Theta_e}\mathbf{C}_1$ and $\tilde{\mathbf{Z}}_3^{\Theta_{FC2}} = \tilde{\mathbf{Z}}_3^{\Theta_{FC1}}\mathbf{C}_2$. Indeed, Θ_{FC1} and Θ_{FC2} correspond to the coefficient matrices $\mathbf{C}_1$ and $\mathbf{C}_2$. Herein, the Frobenius norm is used for coefficient matrices to promote efficient structure mining. Unlike the existing double self-representations [84], it uses the reconstructed feature instead of coefficient to conduct another self-representation, thus avoiding expression bias caused by the differences in statistical characteristics of different domains.

Through the first self-expression, a meaningful coefficient matrix $\mathbf{C}_1$ is derived. Meanwhile, a favorable reconstructed feature $\tilde{\mathbf{Z}}_3^{\Theta_{FC1}}$ is induced, which captures the important hidden structural information and is prone to be ignored. Therefore, it is necessary to explore its self-expression property. Through a following self-expression, the potential of the reconstructed feature can be fully exerted, with salient pixel adjacencies strengthened and omitted pixel relationships replenished.

To integrate the above two parts into a unified constraint, a valid loss is constructed for the DSR module to guide the DSR process of the model, as in the following equation:

$$loss_{DSR} = \beta\left(\|\tilde{\mathbf{Z}}_3^{\Theta_e} - \tilde{\mathbf{Z}}_3^{\Theta_e}\mathbf{C}_1\|_F^2 + \|\tilde{\mathbf{Z}}_3^{\Theta_{FC1}} - \tilde{\mathbf{Z}}_3^{\Theta_{FC1}}\mathbf{C}_2\|_F^2\right) + \gamma\left(\|\mathbf{C}_1\|_F^2 + \|\mathbf{C}_2\|_F^2\right) \tag{9}$$

In this way, the DSR module effectively compensates for the adjacency loss in the single self-expression and fully exploits the latent pixel relationships, thereby performing an accurate self-expression, which directly contributes to the improvement of clustering performance. It should be noted that the above dual self-expressions can be extended to any number of times by adding more self-expression layers, which may be profitable for clustering. However, due to the large-sized coefficient matrices, the sharp increment of computational burden and memory consumption would make the model unaffordable. Hence, two self-expression layers are adopted here.

C. Multilevel Joint Regularizations

As is well known, multilevel features play different roles in target description and have diverse emphases, with shallow features focusing on visual characteristics and deep features concerning semantic properties. In addition, for the DSC model, the expressivity and discriminability of features have a direct impact on the quality and precision of self-representation learning. However, most of the existing DSC methods only focus on the deep-seated features and solely use the deepest encoding feature for self-representation learning, neglecting the importance and complementarity of shallow features, which inevitably leads to certain information loss and performance degradation. Therefore, it is of utmost importance to find a more reasonable way for feature exploration.

At this point, an innovative multilevel feature joint self-representation scheme is developed for the DSC framework to integrate hierarchical complementarity. Considering the limited discriminability of primary convolutional feature, i.e., $\mathbf{Z}_1^{\Theta_e}$, the middle-level and high-level encoding features are fed into the DSR module for joint self-representations through a residual reconnection, as formulated in the following equation:

$$argmin_{\mathbf{C}_j^i}\sum_{i=2}^{3}\|\mathbf{C}_j^i\|_F^2 \; s.t. \; \tilde{\mathbf{Z}}_j^i = \tilde{\mathbf{Z}}_j^i\mathbf{C}_j^i + \mathbf{E}_j^i, \; j = 1, 2 \tag{10}$$

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where $\mathbf{C}_j^i \in \mathbb{R}^{n\times n}$ denotes the coefficient matrix learned by the j-th self-expression based on the flattened feature of the i-th layer, i.e., $\tilde{\mathbf{Z}}_j^i \in \mathbb{R}^{k_i s^2 \times n}$. However, the above self-representation processes are independent of each other, lacking necessary internal relations. Indeed, although multilevel features for the same pixel may have different information, they generally tend to select similar atoms in the linear reconstruction and their coefficient matrices should share a similar structural pattern.

Based on this fact, to realize collaborative assistance and mutual promotion in dual self-expressions, forceful MJRs are constructed to promote consistent expression of multilevel features and enforce their coefficient matrices to share a similar pattern. Specifically, MJRs mainly comprise two terms, i.e., similarity constraint and contrastive constraint. First, from the perspective of structural pattern, a similarity constraint is constructed to enhance the consistency between the coefficient matrices of multilevel features, which can be denoted as $\sum_{j=1}^{2} \|\mathbf{C}_j^3 - \mathbf{C}_j^2\|_F^2$. Second, inspired by contrastive learning, features at different levels for the same pixel can be regarded as a pair of positive samples, and their corresponding coefficients should have high consistency. Therefore, the coefficient matrices $\mathbf{C}_j^2$ and $\mathbf{C}_j^3$ should be consistent. Naturally, a contrastive constraint is constructed to further enhance their consistency, which is denoted as $\sum_{j=1}^{2} max(m - \|\mathbf{C}_j^3 - \mathbf{C}_j^2\|_F^2, 0)$, where m is a small constant and is set as 0.8 in practice. By integrating these two regularizations, an MJR loss can be constructed as:

$$loss_{MJR} = \sum_{j=1}^{2} \left[\|\mathbf{C}_j^3 - \mathbf{C}_j^2\|_F^2 + max\left(m - \|\mathbf{C}_j^3 - \mathbf{C}_j^2\|_F^2, 0\right)\right] \tag{11}$$

In addition, the DSR loss in (9) can be extended to a multilevel feature version, as formulated in the following equation:

$$loss_{MDSR} = \sum_{i=2}^{3} \beta \left(\|\tilde{\mathbf{Z}}_i^{\Theta_e} - \tilde{\mathbf{Z}}_i^{\Theta_e}\mathbf{C}_1^i\|_F^2 + \|\tilde{\mathbf{Z}}_i^{\Theta_{FC1}} - \tilde{\mathbf{Z}}_i^{\Theta_{FC1}}\mathbf{C}_2^i\|_F^2\right) + \gamma\left(\|\mathbf{C}_1^i\|_F^2 + \|\mathbf{C}_2^i\|_F^2\right) \tag{12}$$

Through this manner, MJRs can fully leverage the advantages of multilevel features in dual self-expressions and ensure consistent expression, thus generating more informative and indicative coefficient matrices, which can better recover the underlying structural pattern of HSIs.

D. Weighted Residual Fusion Module

Considering that multiple coefficient matrices generally have differentiated information while sharing a similar structural pattern, it is critical but difficult to precisely fuse them to generate a more complete and clustering-friendly coefficient matrix. Given the diverse contributions of different coefficient matrices to the final clustering performance, it is reasonable to assign different weights to them. Therefore, how to select proper weights has become the key point. In this context, a valid WRF module is put forward for pixel affinity fusion, which adaptively learns the regarding weight for each coefficient matrix according to its significance based on a channel attention and reserves detailed relationships through a residual connection to precisely integrate the differentiated information and reduce affinity loss as much as possible.

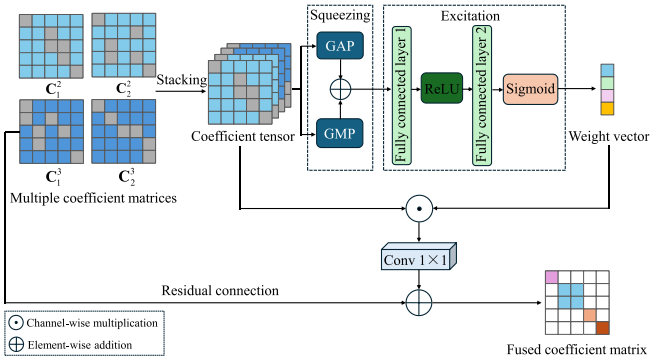


Fig. 3. Structure of the WRF module.

The structure of the WRF module is illustrated in Fig. 3. Specifically, the learned multiple self-representation coefficient matrices $\{\mathbf{C}_1^2, \mathbf{C}_2^2, \mathbf{C}_1^3, \mathbf{C}_2^3\}$ are stacked along the channel dimension to obtain an integrated coefficient tensor $\mathbf{C}_s \in \mathbb{R}^{n\times n\times 4}$. Inspired by the squeezing-excitation (SE) attention [88], $\mathbf{C}_s$ is first squeezed to generate a comprehensive channel vector for weight learning. To enrich feature expressions, two different global pooling operations, i.e., global average pooling (GAP) and global maximum pooling (GMP), are used to compress the spatial information to generate differentiated channel vectors $\mathbf{c}_A \in \mathbb{R}^4$ and $\mathbf{c}_M \in \mathbb{R}^4$. This is because GAP pays more attention to the overall characteristics, while GMP highlights the most significant local part, which can be formulated as:

$$\mathbf{c}_A = GAP(\mathbf{C}_s) = \frac{1}{n\times n}\sum_{p=1}^{n}\sum_{q=1}^{n}\mathbf{C}_{s_{pq}} \tag{13}$$

$$\mathbf{c}_M = GMP(\mathbf{C}_s) = max\left(\mathbf{C}_{s_{pq}}\right) \tag{14}$$

Then, $\mathbf{c}_A$ and $\mathbf{c}_M$ are fused to serve as the basis for weight allocation. Subsequently, an excitation operation is conducted to explore the interactions between different channels to learn the corresponding weight for each channel, which is realized by two FC layers. The first FC layer responses for dimension upgrading, with the second FC layer employed for dimension restoration, which can be expressed as in the following equation:

$$\mathbf{w} = \sigma\left(f_2\left(\delta\left(f_1\left(\mathbf{c}_A \oplus \mathbf{c}_M\right)\right)\right)\right) \tag{15}$$

where $\mathbf{w} \in \mathbb{R}^4$ denotes the weight vector, whose value ranges from 0 to 1, f_1 and f_2 stand for the first and second FC layers, δ and σ represent the ReLU and Sigmoid function, $\oplus$ is an element-wise addition. Based on the learned weights, a refined coefficient tensor $\tilde{\mathbf{C}}_s \in \mathbb{R}^{n\times n\times 4}$ can be induced as:

$$\tilde{\mathbf{C}}_s = \mathbf{C}_s \odot \mathbf{w} \tag{16}$$

where $\odot$ is a channel-wise multiplication operation. Next, a 1×1 convolution is used to integrate the channels of $\tilde{\mathbf{C}}_s$ to yield a high-quality coefficient matrix $\mathbf{C}_F \in \mathbb{R}^{n\times n}$, which adaptively fuses differentiated information of multiple coefficient matrices.

In addition, since pooling operations often lead to detail loss, a residual connection is constructed to integrate the

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detailed pixel adjacencies in the most informative coefficient matrix $\mathbf{C}_1^3$, which is the basis of the existing DSC models, into $\mathbf{C}_F$ to further enhance affinity expressivity and minimize information loss, thus producing a more complete and clustering-friendly coefficient matrix $\mathbf{C} \in \mathbb{R}^{n\times n}$, as in the following equation:

$$\mathbf{C} = \mathbf{C}_1^3 \oplus \mathbf{C}_F \tag{17}$$

By this means, the WRF module can adaptively fuse the complementarity of multiple coefficient matrices to induce more accurate and favorable pixel affinities, which brings about clear gains in the final clustering accuracy.

E. Multifaceted Joint Loss Function

The loss function is another crux for a deep-learning model. Generally, a joint loss can improve the learning ability and stability of the model, especially for multitask optimization. Hence, by comprehensively considering each part of the model, a new multifaceted joint loss function is constructed for the DDSC-MJR model to guide the whole network training.

Apart from data reconstruction loss, DSR loss and MJR loss, similar to [74], [75], [76], and [77], a GR loss is also taken into account to enhance the mining of geometric information in the non-Euclidean space, as formulated in the following equation:

$$loss_G = \varphi tr\left(\mathbf{C}_L^T \mathbf{L} \mathbf{C}_L\right) \tag{18}$$

where $\mathbf{C}_L \in \mathbb{R}^{n\times n}$ is the sum of multiple coefficient matrices, which is defined as $\mathbf{C}_L = \mathbf{C}_1^2 + \mathbf{C}_2^2 + \mathbf{C}_1^3 + \mathbf{C}_2^3$, $tr(\cdot)$ is the trace operation, φ is a regularization parameter, and $\mathbf{L} \in \mathbb{R}^{n\times n}$ is the graph Laplacian matrix of the initial data $\mathbf{X}$. Specifically, the data $\mathbf{X}$ can be constructed as a graph $G = (V, E)$, where V and E denote the node set and edge set. Then, $\mathbf{L}$ can be induced by $\mathbf{L} = \mathbf{D} - \mathbf{A}$, where $\mathbf{A} \in \mathbb{R}^{n\times n}$ is a weight matrix, with each element $\mathbf{A}_{pq} = \exp(-\|\mathbf{X}_p - \mathbf{X}_q\|_2^2/\theta)$ being the corresponding edge weight. Herein, θ is a bandwidth parameter. $\mathbf{D} \in \mathrm{R}^{n\times n}$ is a diagonal matrix of $\mathbf{A}$, with each element defined as $\mathbf{D}_{pp} = \sum_q \mathbf{A}_{pq}$.

Ultimately, by integrating all these individual losses, the multifaceted joint loss function can be expressed as in the following equation:

$$loss = loss_{CAE} + loss_{MDSR} + loss_{MJR} + loss_G \tag{19}$$

With the help of the multifaceted joint loss, each part of the network can be effectively regularized in the training process, which leads to more precise and expressive pixel affinities.

IV. Experimental Results and Analysis

In this section, extensive experiments were conducted on three well-known public HSI datasets to verify the effectiveness of the DDSC-MJR model, with a comprehensive exploration of the results and detailed analysis given as well.

A. Experimental Setting

1) Data Description: For performance validation, three widely used real HSIs were selected, including Indian Pines, Pavia University, and Houston University images, which were acquired by different sensors and have distinct ground object compositions with diverse styles.

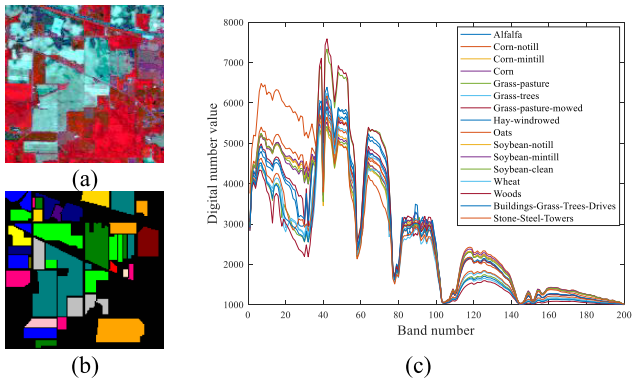


Fig. 4. Indian Pines dataset: (a) False color image, (b) ground truth, and (c) mean spectra of ground objects.

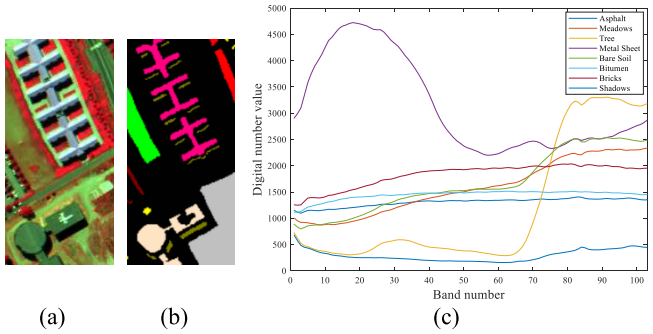


Fig. 5. Pavia University dataset: (a) False color image, (b) ground truth, and (c) mean spectra of ground objects.

The first experimental data is the Indian Pines image, which is a typical agricultural scene acquired by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor over the northwestern Indiana, USA. It has a spatial size of 145×145 and a resolution of 20 m, with 200 bands reserved after abandoning 24 bad bands. There are 16 classes in total, including many similar subclasses with mixed spectra, as depicted in Fig. 4, making this data extremely challenging. For this data, the complete scene was used for clustering analysis.

The second data is the Pavia University image. Different from the first data, it is a typical urban scene collected by the reflective optics system imaging spectrometer (ROSIS) over the campus of Pavia University, Italy. It has a much larger spatial size of 610×340 and a higher resolution of 1.3 m, with 103 bands reserved. It has nine classes in total. For computational efficiency, following most works [70], [71], [72], [73], [74], [75], [76], [77], [78], a typical sub scene located at [151–350, 101–200] was used for clustering analysis, with eight main classes included, as described in Fig. 5.

The third data is the Houston University image, which was offered by the IEEE GRSS Data Fusion Contest in 2013 and was gathered by the National Center for Airborne Laser Mapping (NCALM) sensor over the campus of Houston University and its nearby urban area, USA. It has a very large size of 349×1905, with 144 bands reserved. This scene has

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TABLE I
CONFIGURATIONS FOR THE CAE OF THE DDSC-MJR MODEL

Network part	Layer No.	Operation	Kernel size	Kernel number	Stride size	Padding	Activation
Encoder	1	2D Conv	3×3	24	1×1	Same	BN + ReLU
	2	2D Conv	3×3	24	1×1	Same	BN + ReLU
	3	2D Conv	3×3	32	1×1	Same	BN + ReLU
Decoder	1	2D Deconv	3×3	32	1×1	Same	BN + ReLU
	2	2D Deconv	3×3	24	1×1	Same	BN + ReLU
	3	2D Deconv	3×3	24	1×1	Same	BN + ReLU

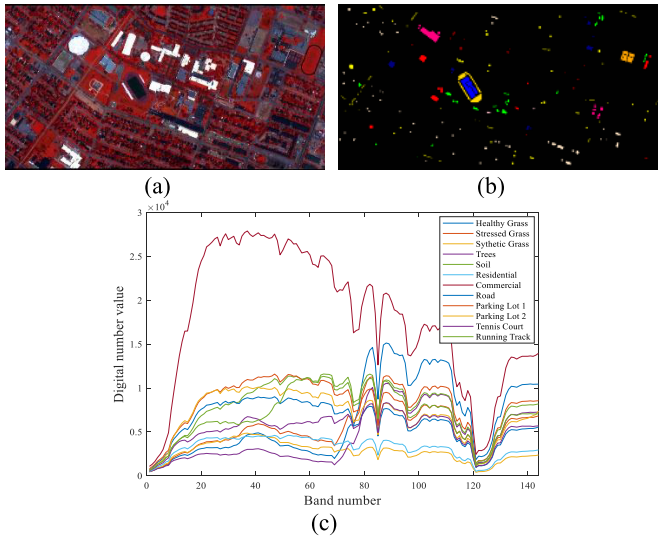


Fig. 6. Houston University dataset: (a) False color image, (b) ground truth, and (c) mean spectra of ground objects.

15 classes, with scattered and complex distribution. Similarly, a typical sub scene located at [1–349, 1–680] was used for clustering [68], [74], [75], [76], with 12 main classes included, as shown in Fig. 6.

2) Evaluation Metrics: The clustering performance was comprehensively evaluated from both visual and quantitative perspectives. Following most works [70], [71], [72], [73], [74], [75], [76], [77], [78], [79], [80], [81], [82], [83], four common metrics were used for accuracy analysis, including producer's accuracy (PA), overall accuracy (OA), normalized mutual information (NMI) and kappa coefficient. The first one refers to the precision of a certain class, with the others referring to the total precision. All these metrics range from 0 to 1, with a larger value indicating a better effect. Because of the lack of real semantics in the clustering results, the Hungarian algorithm was used to automatically permute the clustering label to make it consistent with the ground truth [67].

3) Comparative Methods: To verify the effectiveness and superiority of the proposed DDSC-MJR model, eight representative and SOTA benchmarks were selected for comparative analysis, with both subspace clustering and DSC methods used. Specifically, the baselines include two classical subspace clustering methods, i.e., low rank representation (LRR) [89], efficient dense subspace clustering (EDSC) [90], three popular advanced subspace clustering methods, i.e., efficient graph convolutional subspace clustering (EGCSC), efficient kernel graph convolutional subspace clustering (EKGCSC) [34], graph convolutional optimal transport (GCOT) [53], and three representative SOTA DSC methods, i.e., graph regulated residual subspace clustering network (GR_RSCNET) [74], HyAE [76], and unfolding ADMM for enhanced subspace clustering (UNADMM) [83]. Notably, all benchmarks were implemented by the source codes, with parameters set according to the recommendations.

4) Implementation Details: The DDSC-MJR model is an unsupervised framework based on the CAE architecture, which is trained from scratch, with the detailed configurations given in Table I. In addition, it was implemented using Python language within the TensorFlow (version 2.0.0) framework and was trained by the Adam optimizer with default parameters. The learning rate was set as 0.0002 for all three datasets, with the maximum epoch set as 100, 100, and 300 for Indian Pines, Pavia University, and Houston University images. The hyperparameters in the multifaceted joint loss function were uniformly set as $\alpha = 1$, $\beta = 100$, $\gamma = 0.001$, and $\varphi = 0.001$. All experiments were conducted on a server with an Inter Xeon W-2245 CPU and RAM of 64 GB. Given that several traditional subspace clustering methods were included, for fair comparison, all methods were executed in the same environment on the CPU platform, although the DSC models can be greatly accelerated by the GPU.

B. Experimental Results

The clustering performance of different methods was tested on three real HSI datasets, with both visual maps and quantitative evaluations given for comprehensive comparison.

1) Experiment on the Indian Pines Dataset: The clustering results of different methods for the Indian Pines image are provided in Fig. 7 and Table II. In the table, the improvement amplitude (↑) of the DDSC-MJR model over the second-best model is listed. As illustrated, DSC methods outperformed most subspace clustering methods, and the proposed DDSC-MJR model significantly surpassed the other methods to achieve the best clustering result on this data, with obvious improvements in visual effect and precision. Notably, it is the only one with an OA breaking through 70% for this complex scene, while the OAs of most other arts are under 60%, which demonstrates its effectiveness and superiority.

Specifically, the two classical subspace clustering methods, i.e., LRR and EDSC, failed to distinguish highly similar ground objects due to lacking effective spatial assistance,

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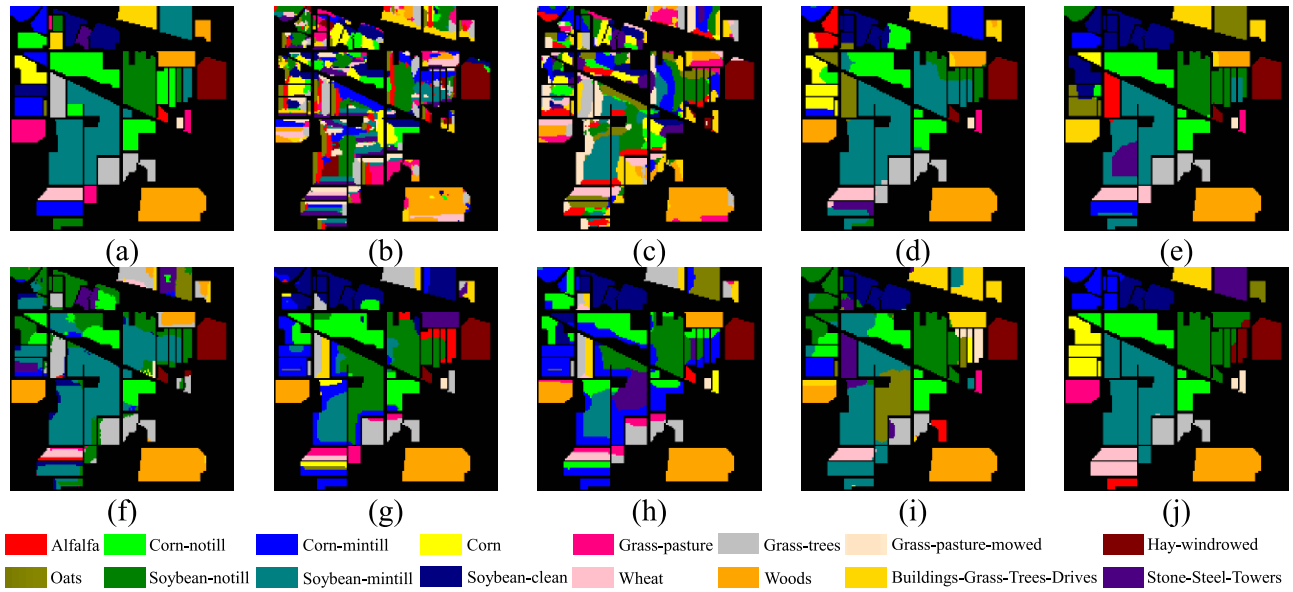


Fig. 7. Clustering maps of different methods on the Indian Pines dataset: (a) Ground truth, (b) LRR, (c) EDSC, (d) EGCSC, (e) EKGCSC, (f) GCOT, (g) GR_RECNET, (h) HyAE, (i) UNADMM, and (j) DDSC-MJR.

TABLE II
CLUSTERING ACCURACY OF DIFFERENT METHODS ON INDIAN PINES DATASET

Metric	Class	LRR	EDSC	EGCSC	EKGCSC	GCOT	GR_RSC NET	HyAE	UNADM M	DDSC-MJR	↑
PA (%)	1	41.30	2.17	0	0	0	0	**100**	0	0	0
	2	15.97	15.13	59.31	**64.78**	35.50	59.38	50.84	44.26	64.22	-0.56
	3	16.63	23.98	26.63	27.23	8.55	43.49	**64.70**	26.02	31.93	4.70
	4	28.69	20.68	63.29	0	0	0	0	0	**100**	100
	5	18.01	14.70	12.22	12.22	0.41	18.22	15.53	12.22	**65.84**	53.62
	6	32.47	31.64	63.01	63.01	**88.49**	73.01	66.30	44.11	63.01	0
	7	0	10.71	**100**	**100**	0	**100**	**100**	0	**100**	0
	8	88.28	97.28	**100**	**100**	**100**	**100**	**100**	**100**	**100**	0
	9	65.00	0	0	**100**	0	0	0	**100**	0	-100
	10	34.67	26.85	19.75	77.67	32.10	64.51	66.67	76.23	**79.94**	2.27
	11	12.14	27.05	76.54	66.19	71.24	21.67	18.33	49.12	**77.03**	10.84
	12	33.39	17.54	42.50	**88.03**	2.02	63.07	75.21	72.68	70.49	-17.54
	13	33.66	61.46	**100**	**100**	60.00	40.98	65.37	99.02	**100**	0
	14	56.92	69.17	89.17	87.43	89.01	71.78	**90.83**	71.46	71.46	-15.97
	15	39.12	36.01	**76.94**	**76.94**	8.55	15.28	23.32	75.13	**76.94**	0
	16	0	48.39	0	0	**100**	0	0	0	0	0
OA (%)		29.12	33.65	60.45	65.43	50.27	48.00	51.62	53.68	**70.20**	4.77
NMI (%)		24.05	36.33	65.71	70.62	51.80	55.60	59.21	60.31	**71.85**	1.23
Kappa		0.2842	0.2797	0.5477	0.6149	0.4244	0.4275	0.4691	0.4825	**0.6650**	0.0501

resulting in serious misclassifications and poor visual effects. For example, the confusion between Corn-notill, Corn-mintill, and Corn was serious, with similar situation existing between Soybean-notill, Soybean-mintill, and Soybean-clean. For over half of the classes, the precision was less than 40%. As a result, these two methods obtained low OAs of only 29.12% and 33.65%. Comparatively speaking, EGCSC improved the clustering performance to a large degree to yield a much smoother map with less misclassifications, by introducing GCR to explore the latent geometric structure of HSIs and integrating nonlocal relationships to enhance model robustness. For most classes, the precision was obviously improved. Consequently, it got a much higher OA of 60.45%. By relieving the linear inseparable problem via kernel mapping, EKGCSC further improved the clustering accuracy. For the Oats class, the PA was improved from 0% to 100%, while for the Soybean-notill and Soybean-clean classes, the gains were also obvious, which indicates the importance of nonlinearity modeling. However, due to the limited generalizability of kernels, it failed to identify Corn and performed poorly for Grass-pasture. By

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introducing an optimal transport strategy, GCOT performed competitively on this dataset. It obtained noticeable precision for Grass-trees and Stone-Steel-Towers. However, limited by the weakness for some classes like Grass-pasture and Soybean-clean, it obtained an OA of only 50.27%.

By introducing deep learning to mine robust feature expressions and nonlinear pixel affinities, the DSC models presented a large potential for HSI clustering. Based on a CAE with residual connections to extract expressive features for self-representation, GR_RSCNET obtained a much better result than LRR and EDSC, to yield a smoother map with obvious improvements for Corn-mintill and Grass-trees. However, due to insufficient exploration of the latent discriminability and affinity, it lost efficacy for highly similar classes, with an OA of only 48.00%. By employing a hypergraph to model higher-order relationships, HyAE obtained a higher OA of 51.62%. For instance, it performed extremely well for Alfalfa. For Corn-mintill and Woods, the PAs were improved from 43.49% and 71.78% to 64.70% and 90.83%, respectively. By using unfold ADMM to deeply optimize the self-representation coefficient matrix, UNADMM further improved the clustering performance to yield a more accurate map. For Wheat and Buildings-Tree-Grass-Drives, it improved the PAs to a large degree, but with reduced PAs for some other classes. Generally speaking, restricted by the common shortcomings in feature mining and affinity learning, these DSC models have limited adaptability to this complex scene, with many misclassifications in the clustering map, which are far from satisfactory.

In contrast, the DDSC-MJR model effectively overcame the above obstacles to achieve a much better clustering result for this scene, by fully leveraging the advantages of hierarchical expressions with MJRs and deeply mining the intrinsic pixel affinities through DSRs. It attained the highest OA of 70.20%, NMI of 71.85%, and Kappa of 0.6650, with noteworthy improvements over the comparative methods. Compared with the second and third best methods, it obtained notable improvements of nearly 5% and 10% in terms of OA. While for the other metrics, i.e., NMI and Kappa, the gains were also significant. Specifically, compared with the second-best model, DDSC-MJR effectively improved the PAs of many classes to a large degree. Especially for Corn, Grass-pasture, and Soybean-mintill, significant gains of 100%, 53.62%, and 10.84% were achieved. However, for some other classes such as Oats, Soybean-clean, and Woods, the disadvantages were also obvious. Overall, DDSC-MJR achieved an impressive performance on this dataset. As noticed, it obtained huge improvements of more than 16%, 18%, and 22% in terms of OA over UNADMM, HyAE, and GR_RSCNET, respectively, which clearly illustrates its superiority. In addition, it obtained the optimal or suboptimal PAs for most classes. It successfully distinguished most similar classes, with high-level precision. Especially for Corn, it obtained a PA of 100%, while most methods failed to identify it. More intuitively, it achieved a more accurate clustering map, which is closest to the ground truth. It effectively reduced misclassifications and smoothed the within class noise, with an ameliorated visual effect.

In spite of the above advances, DDSC-MJR still faced with some limitations. Particularly, for the undominant classes with small samples like Alfalfa, Oats, and Stone-Steel-Towers, it lost efficacy like most other methods. The reason can be that DDSC-MJR failed to capture the intrinsic pixel affinities of these specific classes in self-representation learning, with their coefficients losing category indication. In addition, it exhibited certain weakness in some subclass identification, like Corn and Soybean-clean, due to the strong interference from highly similar classes to self-representation, which degrades model discriminability to some degree. All of these problems will be solved in our future work.

2) Experiment on the Pavia University Dataset: The clustering performance of different methods was further compared on the Pavia University dataset, with the results shown in Fig. 8 and Table III. Similarly, the improvement amplitude of DDSC-MJR over the suboptimal model is given in the table. As can be seen, the clustering results were basically consistent with the first experiment. The DSC methods showed a large superiority to most subspace clustering methods and DDSC-MJR achieved the best result, with the optimal visual effect and highest precision. Because of the distinct distribution of ground objects, most methods obtained high-level precision on this dataset. Nevertheless, DDSC-MJR exhibited clear gains over the other methods to yield an accurate result with tiny errors, which further illustrates its efficacy.

Specifically, LRR and EDSC performed poorly on this data, with serious misclassifications. For example, these two models lost efficacy for Asphalt, Tree, and Bricks, with the PAs being lower than 20%. Consequently, they obtained low OAs of 46.04% and 45.69%. EGCSC greatly reduced misclassifications to yield a smoother map with a much higher OA of 84.42%. For Metal Sheet, the PA was improved from 25.32% and 34.75% to 100%, and for Bare Soil, the gain was also obvious. However, it was less effective for the above three classes. EKGCSC and GCOT further improved the clustering performance to a large degree from the perspectives of nonlinearity modeling and optimal transport, respectively. However, limited by the weak discriminability of shallow features, these two methods still had obvious defects. For example, EKGCSC failed to identify Bricks, and GCOT could not work for the above three classes, which degrades their applicability to some extent.

In contrast, the three DSC models performed competitively on this dataset to yield noticeable OAs around 95% and showed obvious advantages over most traditional models, which indicates the necessity of combining deep learning with subspace clustering. GR_RSCNET effectively distinguished the above three classes to obtain high PAs of 100%. However, it was feeble for Bitumen and Shadows. HyAE and UNADMM performed well for Bitumen by introducing hypergraph and unfold ADMM, respectively, but lost efficacy for Bricks. Consequently, these models failed to yield satisfactory results.

Comparatively, DDSC-MJR dealt well with these defects to yield an impressive result that almost completely masters this dataset. It achieved the highest OA of 99.81%, NMI of 99.50%, and Kappa of 0.9976, with the best visual map that is very close to the ground truth. It attained large improvements of more than 4% in OA over the other three DSC models,

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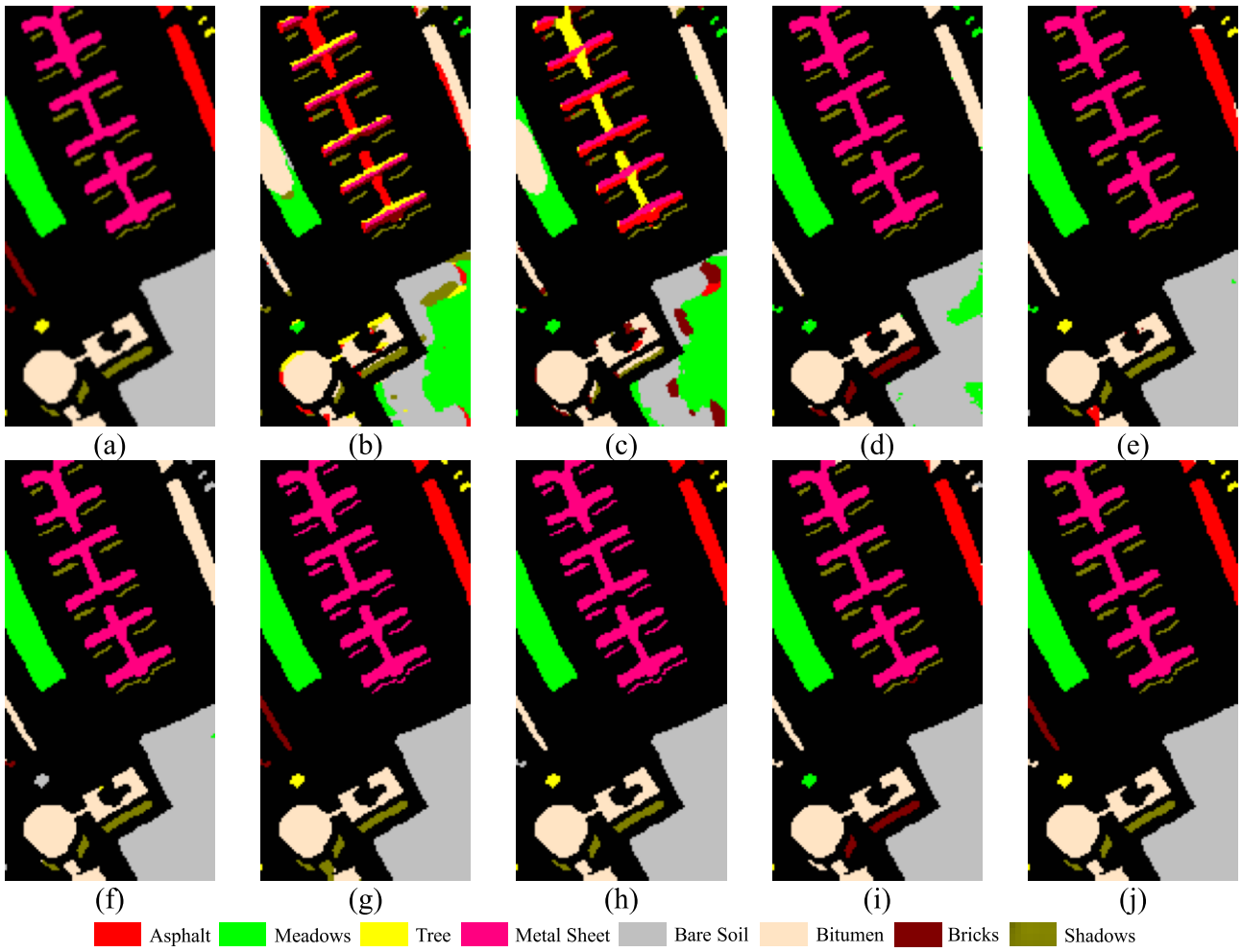


Fig. 8. Clustering maps of different methods on the Pavia University dataset: (a) Ground truth, (b) LRR, (c) EDSC, (d) EGCSC, (e) EKGCSC, (f) GCOT, (g) GR_RECNET, (h) HyAE, (i) UNADMM, and (j) DDSC-MJR.

TABLE III
CLUSTERING ACCURACY OF DIFFERENT METHODS ON THE PAVIA UNIVERSITY DATASET

Metric	Class	LRR	EDSC	EGCSC	EKGCSC	GCOT	GR_RSC NET	HyAE	UNADM M	DDSC-MJR	↑
PA (%)	1	20.00	0.47	0	94.12	0	**100**	**100**	95.06	99.76	5.64
	2	51.04	54.17	**100**	**100**	**100**	**100**	**100**	**100**	**100**	0
	3	11.11	0	4.76	87.30	0	**100**	92.06	4.76	**100**	12.70
	4	25.32	34.75	**100**	**100**	**100**	**100**	**100**	99.70	**100**	0
	5	44.31	39.66	89.29	99.88	99.88	**100**	**100**	**100**	**100**	0.12
	6	81.74	93.49	99.53	95.23	99.65	94.88	**100**	**100**	**100**	4.77
	7	0	12.77	0	1.06	10.64	**100**	0	0	88.30	87.24
	8	86.70	66.20	59.28	**100**	99.72	40.44	40.44	57.89	**100**	0
OA (%)		46.04	45.69	84.42	97.33	91.02	95.58	95.13	94.86	**99.81**	2.48
NMI (%)		52.47	56.91	84.01	96.49	92.33	94.59	93.88	93.90	**99.50**	3.01
Kappa		0.3617	0.3586	0.7968	0.9521	0.8804	0.9468	0.9354	0.9323	**0.9976**	0.0455

with the help of MJRs and DSRs. A gain of about 2.5% was obtained in OA compared to the second-best EKGCSC, with more significant gains obtained over the other traditional models. As noticed, it achieved large improvements for the vast majority of classes, without any negative effect. Moreover, it presented a powerful ability for each class with few misclassifications, with a remarkable precision of 100% for most classes. In general, DDSC-MJR achieved an excellent performance for this scene. Nevertheless, DDSC-MJR only achieved the suboptimal PAs for Asphalt and Bricks with

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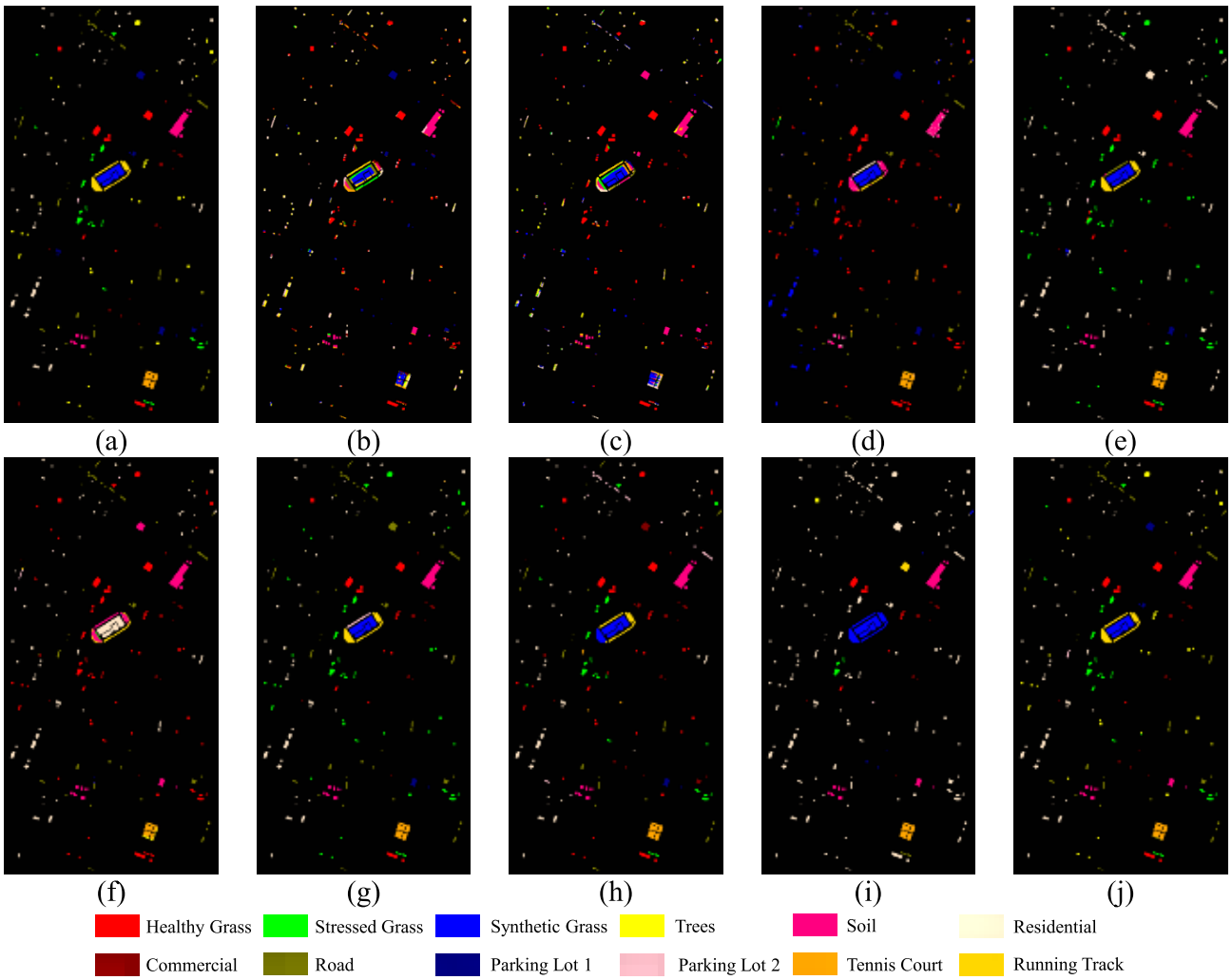


Fig. 9. Clustering maps of different methods on the Houston University dataset: (a) Ground truth, (b) LRR, (c) EDSC, (d) EGCSC, (e) EKGCSC, (f) GCOT, (g) GR_RECNET, (h) HyAE, (i) UNADMM, and (j) DDSC-MJR. (All maps are rotated 90°).

relatively smaller samples, due to the large difficulties in precise atom selection for self-representation of these classes.

3) Experiment on the Houston University Dataset: The last experiment was conducted on the Houston University dataset, with the results given in Fig. 9 and Table IV. As observed, there were significant gaps between DSC and traditional models, while DDSC-MJR presented overwhelming advantages over the other methods. Because of the high similarity and scattered complex distribution of ground objects, the clustering accuracy of most comparative methods was at a low level. DDSC-MJR displayed a strong adaptability to this complex scene to yield an accurate result with the best visual effect and highest precision, far ahead of the other arts, which further indicates its effectiveness and superiority.

In particular, the two classical subspace clustering methods, i.e., LRR and EDSC, performed badly on this dataset, to obtain the lowest OAs of nearly 41%. The three graph convolutional subspace clustering methods delivered better performance, but with limited improvements. For example, the OAs of EGCSC and GCOT were around 53%, with smaller increase rates compared to the former two experiments, which indicates their limited adaptability to the complex scene. Although EKGCSC obtained much better precision, with an OA over 74%. There were still a large number of misclassifications in the clustering map, especially for Trees and Road. By comparison, the DSC methods significantly reduced misclassifications and presented incomparable advantages, except for UNADMM. However, they still lack the strength for Trees, with obvious weakness for Packing Lot 1. As a result, unsatisfactory results were yielded, with limited OAs of only 83.22% and 78.53%.

In contrast, DDSC-MJR addressed these shortcomings well to achieve the best result on this dataset, with OA, NMI, and Kappa reaching 92.96%, 90.98%, and 0.9216, respectively, significantly outperforming the other methods. Compared with the second-best method, i.e., GR_RECNET, an improvement of nearly 10% was obtained in terms of OA. It showed large advantages over GR_RECNET for most classes, with a few exceptions like Healthy Grass, Road, and Tennis Court. While for the other methods, the improvements were more obvious. It obtained the optimal or suboptimal PAs for most classes. Especially for Trees, it obtained a high PA of 98.64% while the other methods failed to identify it. Moreover, the clustering

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TABLE IV
CLUSTERING ACCURACY OF DIFFERENT METHODS ON THE HOUSTON UNIVERSITY DATASET

Metric	Class	LRR	EDSC	EGCSC	EKGCSC	GCOT	GR_RSC NET	HyAE	UNADM M	DDSC-MJR	↑
PA (%)	1	91.77	99.82	**100**	**100**	**100**	**100**	98.35	42.96	95.80	-4.20
	2	9.61	9.80	0	95.38	0	**100**	97.78	20.15	**100**	0
	3	59.40	79.63	99.86	**100**	0	**100**	**100**	**100**	**100**	0
	4	17.65	5.66	0	0	0.45	0	0	0.23	**98.64**	98.64
	5	87.37	84.34	82.89	**100**	**100**	99.34	**100**	**100**	**100**	0.66
	6	25.98	27.24	0.25	98.36	97.10	89.28	98.87	**99.75**	88.90	-0.38
	7	0	67.92	70.44	70.44	70.44	70.44	86.16	64.78	**88.68**	18.24
	8	25.45	20.72	73.25	0	75.53	**99.84**	57.26	0	92.66	-7.18
	9	60.21	1.31	**62.30**	26.44	0	26.44	26.44	0	47.38	20.94
	10	3.85	7.69	0	**53.85**	0	0	0	0	0	0
	11	18.46	16.59	95.79	**100**	74.07	**100**	**100**	**100**	95.56	-4.44
	12	19.70	20.30	17.12	80.45	36.82	80.61	64.39	0	**100**	19.39
OA (%)		41.53	41.05	52.86	74.17	53.14	83.22	78.53	53.31	**92.96**	9.74
NMI (%)		44.87	46.58	69.52	85.22	69.46	87.45	86.83	69.25	**90.98**	3.53
Kappa		0.3558	0.3479	0.4784	0.7105	0.4745	0.8132	0.7624	0.4720	**0.9216**	0.1084

TABLE V
COMPARISON WITH THE OTHER SOTA DEEP MODELS

Methods	Indian Pines dataset			Pavia University dataset			Houston University dataset		
	OA (%)	NMI (%)	Kappa	OA (%)	NMI (%)	Kappa	OA (%)	NMI (%)	Kappa
3D-ACAE				65.49	65.49	\			
CMSCGC	\	\	\	96.69	97.02	0.9565	87.21	**91.68**	0.8580
PSCPC	64.46	64.52	0.6032	\	\	\	\	\	\
S^2DL	64.70	59.10	0.6020	\	\	\	\	\	\
EMVCC	65.13	69.81	0.6184	\	\	\	\	\	\
DDSC-MJR	**70.20**	**71.85**	**0.6650**	**99.81**	**99.50**	**0.9976**	**92.96**	90.98	**0.9216**

map was basically consistent with the ground truth, with distinct semantic and clear distribution, even for small ground objects. However, for some minor classes like Parking Lot 1 and Parking Lot 2, DDSC-MJR suffered from the downgraded performance, as a result of inaccurate self-representation for these classes, which will be our next effort.

C. Comparison With the Other SOTA Deep Models

In order to further validate the effectiveness of the proposed DDSC-MJR model, five recent SOTA deep clustering methods for HSIs were employed for comparison, including 3-D attention CAE (3D-ACAE) [71], contrastive multiview subspace clustering based on graph convolutional networks (CMSCGC) [77], pixel-super pixel contrastive learning and pseudo-label correction (PSCPC) [91], super pixel-based and spatially regularized diffusion learning (S^2DL) [59], and enhanced multiview contrastive clustering (EMVCC) [69]. Among them, both DSC models and other deep framework-based models are included. Since the source codes of these methods are unavailable, the clustering maps cannot be precisely reproduced. Hence, following the common strategy [74], [77], the reported precision on the same dataset in the literatures were used for direct comparison, with the results listed in Table V.

From Table V, similar convolutions can be drawn that DDSC-MJR presented obvious advantages over the other SOTA deep models to achieve the highest precision on all three datasets. Specifically, for the Indian Pines image, it obtained large gains of 5.07%, 2.04%, and 4.66% in terms of OA, NMI, and Kappa, respectively, compared with the second-best method, i.e., EMVCC. For the Pavia University and Houston University images, it obtained clear gains of 3.12% and 5.75% in terms of OA over the second-best method, i.e., CMSCGC. All these results demonstrate the superiority of the DDSC-MJR model.

D. Efficiency Comparison of Different Methods

In this section, the efficiency of different methods was further compared, to analyze the practicality of the DDSC-MJR model. Considering that some traditional models were used as the benchmarks, to ensure the fairness of comparison, all methods were executed in the same environment on the CPU platform, with the time cost of different methods on all three datasets given in Table VI.

From Table VI, it can be clearly observed that DSC models are generally more time-consuming than traditional models, because of their complicated feature engineering. Specifically, EDSC and EGCSC are very efficient, while EKGCSC and GCOT have higher time costs, with LRR being an exception, which is time-consuming due to the high computational

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TABLE VI
TIME COST OF DIFFERENT METHODS IN CPU (S)

Methods	IP dataset	PU dataset	HU dataset
LRR	16953.84	3548.78	3175.71
EDSC	**53.26**	53.57	67.25
EGCSC	234.63	**52.77**	**60.23**
EKGCSC	485.25	92.53	155.99
GCOT	232.53	65.47	92.91
GR_RSCNET	4643.08	1954.29	1736.72
HyAE	5683.70	5468.07	3201.47
UNADMM	14809.08	5842.37	5207.95
DDSC-MJR	13230.55	2941.70	2755.87

IP: Indian Pines; PU: Pavia University; HU: Houston University

complexity caused by the large-scale rank optimization of the coefficient matrix. GR_RSCNET is most efficient among the four DSC models, although its time cost is much higher than that of most traditional models. HyAE consumes more time on all three datasets because it comprises a more complex structure when introducing an additional hypergraph structure mining module to improve feature discriminability. UNADMM is most time-consuming, especially for the Indian Pines image, due to its complex coefficient matrix optimization process. By comparison, our method has compromised efficiency, with a slightly higher time cost than GR_RSCNET on the Pavia University and Houston University datasets, but is more efficient than HyAE and UNADMM. Although it is relatively time-consuming for the Indian Pines image, as a result of the introduction of MJRs, DSR and WRF modules, its time cost is lower than that of UNADMM and is at an acceptable level. Nevertheless, DDSC-MJR can be an attractive tool in practice, stemming from its much higher clustering accuracy and acceptable time cost.

E. Ablation Studies

The proposed DDSC-MJR model has three core innovative components, i.e., DSR module, MJRs, and WRF module, which jointly determine the final clustering performance. With the addition of each part, the associated loss is added to the total loss function of the whole network. Apart from these core terms, GR was also introduced to improve model robustness and form the final multifaced joint loss. To validate the efficacy of each part, a series of ablation experiments were conducted on all three datasets. The ECA embedded CAE with single self-expression layer was taken as the baseline, whose loss function is as same as traditional DSC models. Subsequently, the three core components and GR were added step by step. The results are listed in Table VII.

1) Ablation of DSR Module: The DSR module extends traditional single self-expression to consecutive self-expressions, which effectively explores the potential of the reconstructed feature and fully mines the intrinsic relationships between pixels. It determines the quality of the coefficient matrix and has an important influence on the final accuracy. As in Table VII, by adding the DSR module, clear gains can be noticed for the metrics on all three datasets. For example, the OA increased by 3.51%, 1.25%, and 3.07% for Indian Pines, Pavia University, and Houston University images, respectively. Similar gains can be noticed for NMI and Kappa. All these results demonstrate the necessity of the DSR module.

2) Ablation of MJRs: MJRs extend the traditional single-feature self-representation problem based on the deepest feature to a multifeature joint self-representation problem and effectively ensure consistent expression of multilevel features by promoting their coefficient matrices to share a similar structural pattern, thereby integrating hierarchical complementarity to enhance the discriminability of the model. It determines feature richness and has a direct impact on the clustering performance. As illustrated, by adding this part, all metrics were improved to a certain degree on all three datasets. Specifically, the OA increased by 1.60%, 0.73%, and 0.18% for Indian Pines, Pavia University, and Houston University images, respectively. Similarly, clear gains were found for Kappa as well. All of these suggest the importance of MJR.

3) Ablation of the WRF Module: The WRF module integrates the advantages of multiple coefficient matrices in a precise manner by learning adaptive weights and reduces detail loss by residual connection, which decides the completeness and precision of the learned pixel affinities and directly contributes to the improvement of partition accuracy. As can be seen, the WRF module brought about significant increments in the metrics on all three datasets. For instance, it obtained large gains of 3.66%, 2.10%, and 7.71% in OA for Indian Pines, Pavia University, and Houston University images, with obvious gains in NMI and Kappa as well, which indicates the reasonability of the WRF module.

As noticed, although the DSR module and MJRs effectively improve affinity mining and discriminability learning of the model, their simple coefficient fusion strategy based on the element-wise addition makes the complementary information contained in different coefficient matrices unable to be fully integrated. In addition, the redundant information of multiple coefficient matrices inevitably leads to certain bias and negative impacts on affinity expression. As a result, the potential value of the learned multiple coefficient matrices cannot be fully unleashed by the DSC module and MJRs, which limits their clustering performance to some degree. By comparison, the WRF module learns adaptive weights for different coefficient matrices by fully exploring their significances, thereby precisely integrating their complementarity and advantages in the fusion process, thus generating a more expressive and clustering-friendly coefficient matrix for adjacency matrix construction, which effectively overcomes the above obstacles and more accurately describes pixel affinities. Therefore, the WRF module brought about more significant performance improvements compared to the other two components.

4) Ablation of GR: GR effectively mines geometric information in the non-Euclidean space, which further enhances the learning ability and stability of the model. It assists the model to more comprehensively mine the structural information of HSIs. As observed, by adding GR to form the DDSC-MJR model, best results were achieved on all three datasets, with the OA improved by 0.33%, 0.06%, and 0.09%, respectively, which confirms the assistance of GR and the effectiveness of the multifaced joint loss function.

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TABLE VII
ABLATION STUDIES OF THE DDSC-MJR MODEL ON ALL THREE DATASETS

Network components	Indian Pines dataset			Pavia University dataset			Houston University dataset		
	OA (%)	NMI (%)	Kappa	OA (%)	NMI (%)	Kappa	OA (%)	NMI (%)	Kappa
Baseline	61.10	67.85	0.5728	95.67	93.51	0.9427	81.91	86.53	0.7991
+DSR	64.61	69.82	0.6104	96.92	95.44	0.9596	84.98	87.09	0.8327
+DSR+MJR	66.21	68.75	0.6201	97.65	96.62	0.9691	85.16	86.77	0.8347
+DSR +MJR +WRF	69.87	71.56	0.6610	99.75	99.26	0.9967	92.87	**90.99**	0.9206
+DSR+MJR +WRF+GR	**70.20**	**71.85**	**0.6650**	**99.81**	**99.50**	**0.9976**	**92.96**	90.98	**0.9216**

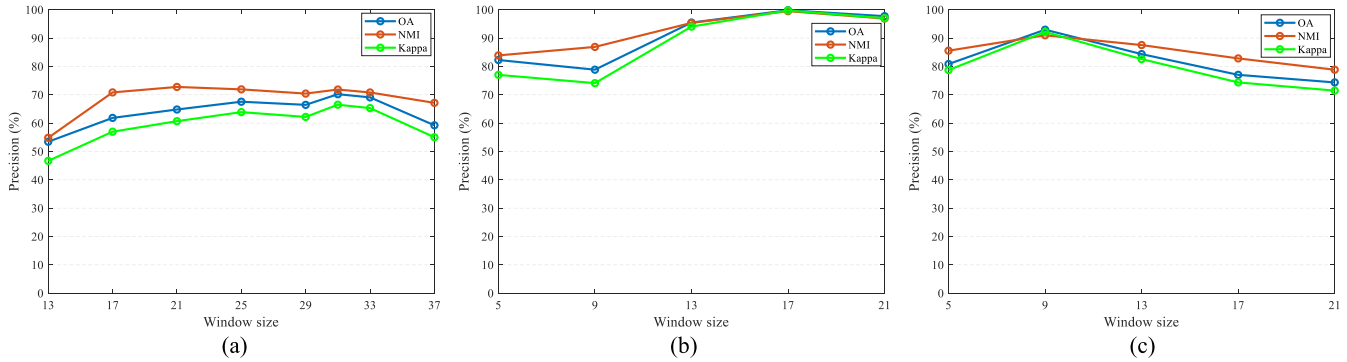


Fig. 10. Sensitivity analysis of window size on all three datasets: (a) Indian Pines dataset, (b) Pavia University dataset, and (c) Houston University dataset.

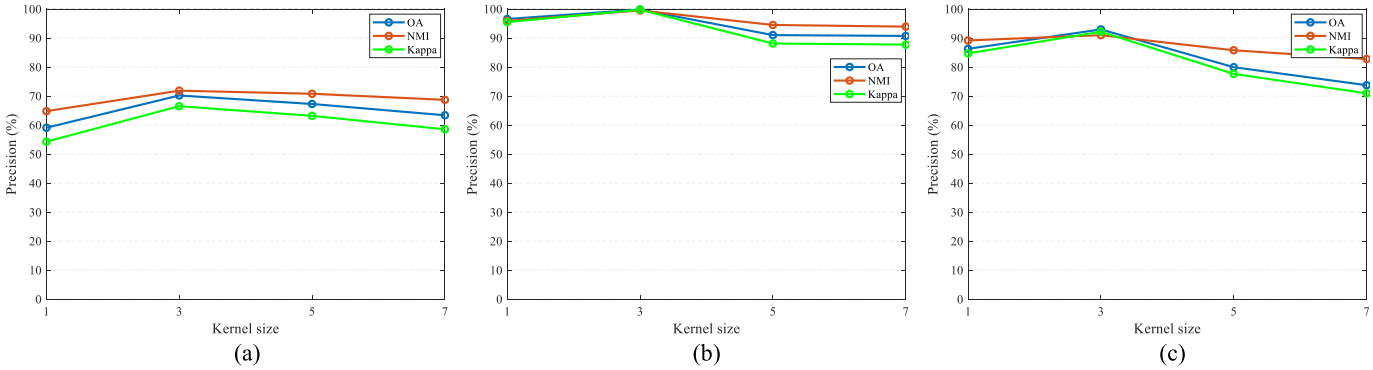


Fig. 11. Sensitivity analysis of kernel size on all three datasets. (a) Indian Pines dataset, (b) Pavia University dataset, and (c) Houston University dataset.

F. Parameter Analysis

In the DDSC-MJR model, there are two key parameters in feature extraction, including window size and kernel size. To explore the influence of these two parameters on the clustering performance, a set of experiments was conducted on all three datasets, with OA, NMI, and Kappa as indicators. The change trend curves are shown in Figs. 10 and 11.

1) Influence of Window Size: To deeply mine discriminative features through convolutions, a centralized window is opened for each pixel. A large window can enrich spatial contents for convolutional learning but may introduce irrelevant interference from other categories, while a small window can ensure the purity of ground objects but with limited context information. Therefore, the window size is important for feature extraction. The change trend curves of precision versus window size are used to indicate its impact, with window size getting its values from the range of [13, 37] for the Indian Pines image, [5, 21] for the Pavia University image and Houston University image, while sharing the same interval of 4 for all three images.

From Fig. 10, it can be observed that the window size has a large impact on the clustering performance, but with a distinct law. For the agricultural scene with an aggregated and smooth distribution of ground objects like the Indian Pines image, the optimal window size is relatively large, i.e., 21. While for the urban scene with many small targets, complex distribution, and sharply changed land objects like the Houston University image, the optimal window size is relatively small, i.e., 9, thereby excluding the interference of surrounding objects from other classes. For the Pavia University image with composite characteristics, the optimal window size is moderate, i.e., 17. In general, the DDSC-MJR model delivers a relatively stable performance in a certain range of window size. All of these give helpful guidance to practical applications.

2) Influence of Kernel Size: The convolutional kernel size determines the range of the receptive filed and the strength of spatial correlations. A small kernel is favorable to learning

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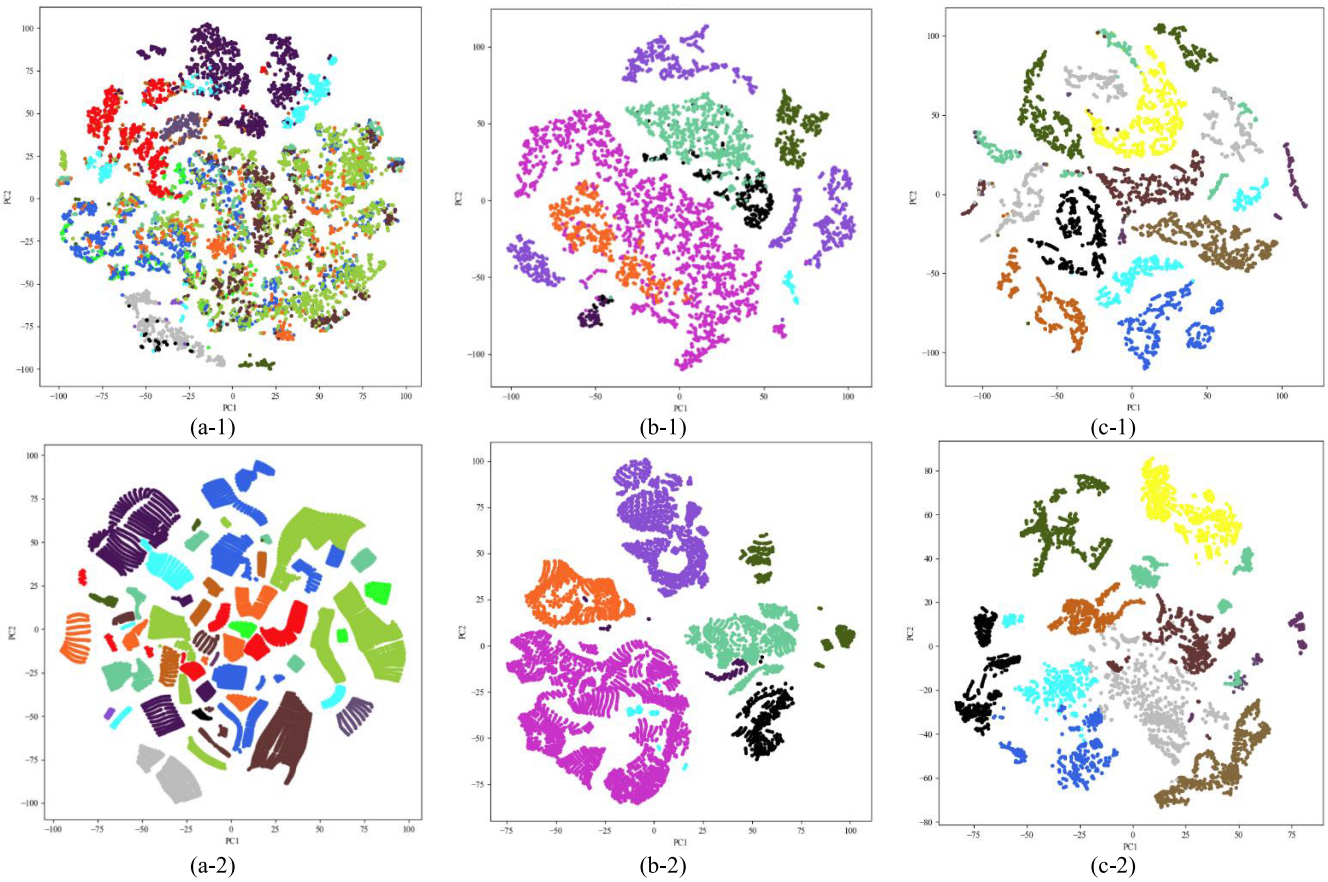


Fig. 12. Visualization of feature space using *t*-SNE: (a-1) original feature space for Indian Pines image, (a-2) deepest feature space learned by DDSC-MJR for Indian Pines image, (b-1) original feature space for Pavia University image, (b-2) deepest feature space learned by DDSC-MJR for Pavia University image, (c-1) original feature space for Houston University image, and (c-2) deepest feature space learned by DDSC-MJR for Houston University image.

local details, while a large kernel is favorable to capturing richer correlations. Hence, the kernel size is of utmost importance and directly determines the expressivity of features. Similarly, the change trend curves of precision versus kernel size are given, with the same value range of [1, 7] and an interval of 2 for all three images.

As shown in Fig. 11, the kernel size affects the clustering performance to a large extent. It can be found that the precision rises first and then falls with the increment of kernel size, with a shared optimal value of 3 for all three datasets. It is because a too small kernel size wastes a lot of neighborhood correlations and leads to certain discriminability loss while a too large kernel size includes many irrelevant information to induce nonrepresentative features. In total, the performance of DDSC-MJR is relatively stable in a certain range.

G. Visualization of Feature Space and Adjacency Matrix

To intuitively analyze the effectiveness of the DDSC-MJR model in feature extraction and affinity learning, the learned deep feature expressions and coefficient matrix are visualized, with the results provided in Figs. 12 and 13.

Discriminative and robust feature expressions are one of the largest advantages of deep learning and are the basis of self-expression, which decides sample distributions in the feature space and plays an important role in clustering. To explore the effectiveness of DDSC-MJR in feature extraction, the deepest feature space induced by the ECA embedded CAE was compared with the original feature space of HSIs, with *t*-SNE used for dimensionality reduction and visualization [10], [78]. Fig. 12 clearly indicates the change in sample distributions after feature engineering. As illustrated, samples of various classes mix up seriously in the initial feature space, with same categorical samples scattering and splitting into several parts, especially for the Indian Pines image, which creates a large challenge for affinity learning. Comparatively, samples in the deepest feature space have neat and concentrated distributions, with samples from the same class aggregated together to form distinct clusters and samples from different classes clearly separated, which better recovers the latent subspace structure and is conductive to affinity learning. In general, DDSC-MJR can effectively learn expressive features for self-expression.

The adjacency matrix captures the latent affinities between pixels, with each element describing the similarity between the corresponding pixel pair, which is induced by the learned coefficient matrix as in (4) and is the basis for pixel partition. Generally, the adjacency matrix is nonnegative and symmetrically distributed, with large elements indicating high similarity between the associated pixels, whose quality has a direct influence on the final clustering accuracy. To explore the efficacy of DDSC-MJR in affinity learning, the adjacency matrix induced by the fused coefficient matrix is visualized. As shown in Fig. 13, adjacency matrices on all three datasets have

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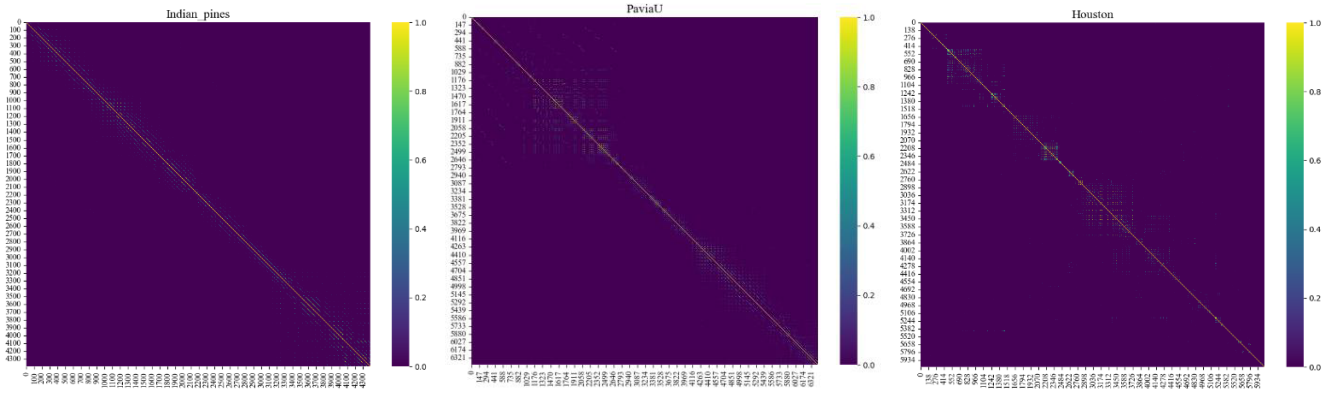


Fig. 13. Visualization of the adjacency matrix produced by DDSC-MJR on all three datasets: (a) Indian Pines dataset, (b) Pavia University dataset, and (c) Houston University dataset.

a sparse distribution, with nonzero elements mainly distributed along the diagonal line and larger values concentrated in its nearby area. As a result, they have an obvious block diagonal structure, with each block corresponding to a certain cluster, which well recovers the underlying clustering structure and makes pixels easily partitioned into the correct clusters. It further confirms the effectiveness of the DDSC-MJR model.

V. Conclusion

Faced with the obstacles of the existing DSC models in latent discriminability and affinity mining, this article proposed a novel DSC framework named DDSC-MJR for HSIs. First, an effective DSR module was developed to explore the hidden value of the reconstructed feature and fully mine the intrinsic pixel affinities by consecutive self-representations. Second, efficacy MJRs were constructed to promote consistent expression of deep and shallow features, thus integrating hierarchical complementarity to further enhance model discriminability. Third, a valid WRF module was designed to fuse the learned coefficient matrices in a precise manner to generate a more complete and clustering-friendly adjacency matrix. Ultimately, a multifaceted joint loss was put forward to simultaneously constrain each part of the network in the training process, with an additional GR introduced to further enhance model robustness. Extensive experimental results on three public HSI datasets illustrate that DDSC-MJR significantly outperforms the other SOTA methods and has a substantial practical potential.

However, there are still some limitations for DDSC-MJR. First, the model only focuses on local feature learning, lacking global expressions, due to the inherent defect of convolutions, which degrades feature discriminability to some extent. In addition, restricted by the large size of coefficient matrices, the model is troubled by a large complexity, with limited scalability to large scenes, which certainly degrades its attractiveness in real applications. All of these will be solved in our next efforts.

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Han Zhai (Member, IEEE) received the B.S. degree in remote sensing science and technology from Shandong University of Science and Technology, Qingdao, China, in 2014, and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2019.

He has been an Associate Professor with the School of Geography and Information Engineering, China University of Geosciences, Wuhan, since 2019. He has published a series of high-quality papers and patents. His research interests include remote sensing intelligent recognition, hyperspectral information processing, artificial intelligence, and remote sensing applications.

Dr. Zhai has been a reviewer of more than ten international journals, such as *ISPRS Journal of Photogrammetry and Remote Sensing*, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, and IEEE GEOSCIENCE AND REMOTE SENSING LETTERS.



Jun Yu (Student Member, IEEE) received the B.S. degree in geographic information science from Central South University, Changsha, China, in 2023. He is currently pursuing the M.S. degree with the School of Geography and Information Engineering, China University of Geosciences, Wuhan, China.

His research interests include deep learning, hyperspectral image processing, and clustering.

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